

Phase 4 Deliverables Checklist

Planned Deliverables → Status

1. Three Functional Smart Contracts → DELIVERED (adapted)

The original plan called for three ink! contracts. The implementation uses a unified architecture:

Table 1: Smart Contract Deliverables

Deliverable	Status	Location
SubscriptionManager (subscription logic)	Delivered	pallets/content-rights/
PayPerView (one-time access)	Delivered	pallets/content-rights/
PurchaseVerification (ownership)	Delivered	pallets/content-rights/
RightsManager ink! 6 contract	Additional	contracts/rights-manager/
pallet-rights-verifier	Additional	pallets/rights-verifier/

2. Integrated Local Testnet Deployment → DELIVERED

Table 2: Testnet Deployment Status

Component	Status	Details
4-chain Zombienet	Passing	Relay + Bridge Hub + AssetHub + Content Rights
Ethereum local network	Passing	Geth + Lodestar + Gateway contracts
HRMP channels	Passing	BH↔AH, AH↔CR
Snowbridge beacon client	Passing	Initialized on Bridge Hub
Relayers	Passing	Beacon relay + Ethereum relay running

3. XCM Implementation → DELIVERED

Table 3: XCM Implementation Status

Component	Status	Details
Cross-chain subscription renewal	Passing	renew_subscription_cross_chain extrinsic
Cross-chain PPV access	Passing	pay_per_view_cross_chain extrinsic
Cross-chain ownership transfer	Passing	transfer_ownership_cross_chain extrinsic
Ethereum bridge (Snowbridge)	Passing	Full v1 local setup with relayers
XCM paid execution	Passing	WithdrawAsset + BuyExecution pattern

4. Unit Test Suite (80%+ coverage) → DELIVERED

Table 4: Test Suite Status

Test Layer	Count	Status
Unit tests (pallet)	23	Passing
XCM simulator tests (Layer 2)	Multiple	Passing
Zombienet E2E tests (Layer 3)	Multiple	Passing

5. Initial Implementation Documentation → DELIVERED

Table 5: Implementation Documentation

Document	Location
Snowbridge setup plan	content-rights-parachain/docs/SNOWBRIDGE_SETUP.md
Snowbridge session log + lessons	content-rights-parachain/docs/SNOWBRIDGE_SESSION_LOG.md
Deploy & call guide	content-rights-parachain/docs/DEPLOY_AND_CALL.md
Implementation decisions	See Current status.md in project root

Additional Deliverables (Beyond Plan)

These were not in the original plan but were needed for the thesis:

- **pallet-rights-verifier** — Cross-chain Merkle storage proof verification
- **Custom pallet-revive precompile** — Efficient contract→pallet calls
- **snowbridge-full-setup.sh** — Automated Ethereum bridge setup script
- **3-layer test architecture** — Unit + XCM simulator + Zombienet E2E
- **Full Snowbridge E2E bridge** — 2 ETH successfully bridged from Ethereum to AssetHub via Gateway → relay → Bridge Hub → AssetHub (verified 2026-03-23)
- **Automatic royalty propagation** — `set_royalty_splits` extrinsic + `pay_with_royalties` distributes payments to up to 10 collaborators with basis-point precision
- **Scheduled auto-renewal** — `on_initialize` hook + `enable_auto_renew/disable_auto_renew` extrinsics process expired subscriptions automatically each block
- **Metadata-carrying XCM** — `query_rights_metadata` emits complete `RightsMetadata` struct (pricing, royalty config, content details) for cross-chain consumers
- **56 unit tests** (23 original + 10 security + 5 royalty + 5 auto-renewal + 2 metadata + 11 XCM)

Phase 5

For Phase 5 testing results (Weeks 17–20), see *Phase 5 — Testing, Evaluation, & Thesis Integration*.